

Analysis of Epidemic Spreading Process in Adaptive Networks

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Abstract—In realistic world, vast majority of networks exhibit layered properties, i.e., the same nodes share different links in each layer (referred as multi-layer networks). Here, the epidemic spreading process on multi-layer networks is investigated, where an infectious yet indistinguishable (as symptoms are not presented) state is considered (i.e., an exposed state). Motivated by the fact that subjects are inclined to take preventive measure: cutting potential connection with infection sources during undesirable infections, we consider the epidemic spreading process with such an adaptive framework: a susceptible/exposed individual will disconnect with infected neighbors probabilistically. Once a connection is removed, one new connection will be constructed with uninfected one. The epidemic spreading process will be greatly affected with the adoption of adaptive strategy. Furthermore, we also study the effect of varying community number via incorporating new individuals, and find the consideration of community movement plays an important role in the epidemic re-occurrence. Various benchmarks are analyzed via using the proposed model as well to investigating corresponding spreading behaviors.

Index Terms—Multilayer networks, adaptive networks, link rewiring, exposed state, required exposed periods, Markov chain.

I. INTRODUCTION

RECENTLY, various studies are performed to obtain detailed information about the evolving of complex networks [1], [2]. The real-world communities usually exhibit layered properties, e.g., individuals might be involved in various social ties (including family, friends, coworkers and etc.) or related via different Internet tools. For this aim, multi-layer networks become a useful tool to mimic these empirical systems and have attracted great research attention, e.g., the European air transport system [3] and power grid system [4]. While multiplex networks refer to a special category of multi-layer networks with exactly the same set of nodes across all layers.

Manuscript received June 14, 2018; revised August 11, 2018 and September 26, 2018; accepted October 16, 2018. Date of publication October 23, 2018; date of current version June 26, 2019. This work was supported in part by the National Science Foundation for Young Scientists of China under Grant 61601371, in part by the Fundamental Research Funds for the Central Universities under Grant 3102018zy034, in part by the Natural Science Basic Research Plan in Shaanxi Province of China under Grant 2018JQ6075, in part by the Aviation Science Fund of China under Grant 2016ZD53035, and in part by the Industry-Academy-Research Cooperation Project of AVIC under Grant cxy2013XGD14. This brief was recommended by Associate Editor M. Small. (Corresponding authors: Yangming Guo; Zhen Wang.)

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Digital Object Identifier 10.1109/TCSII.2018.2877406

Among various applications of multiplex networks, the epidemic spreading process upon them is a special topic attracting broad interests [5]–[10]. In [11]–[14], two-layered multiplex networks (consisting of a virtual layer (i.e., V) and a physical one (i.e., P)) are proposed to analyze the epidemic spreading process. Nodes on different layers represent either physical individuals or virtual agents; connections on the physical layer indicate practical relationships with social ties. On physical layer, the epidemic spreads via exchanging biological elements which carry the virus. Whereas, on the virtual layer, nodes can exchange information about the epidemic, enabling individuals to be aware of the epidemic.

In spite of great progress, some novel infectious diseases, such as H1N1, SARS, have recently caused tremendous threats to human society. In this sense, taking preventive measures, e.g., reducing contact outside, taking the vaccination and so forth becomes very useful [15]–[17], which, however, to some extent incurs the changing of network topology, referred as adaptive networks. In an adaptive network, the topology varies if an uninfected individual disconnects with infected neighbors probabilistically. Once a connection is removed, the individual will connect with another uninfected individual. In [18], with adaptive susceptible–infected–susceptible (ASIS) model, the relationships between spreading behaviors and time-varying topology are thoroughly studied. In [19], a preferentially reconnection strategy is presented for adaptive networks with the adoption of susceptible–infected (SI) model. Moreover, similar mathematical analyses are also performed in adaptive networks [20], [21]. If links between susceptible individuals and infected neighbors are removed permanently, the epidemic will be eliminated rapidly [22]. At variance, the effects of adding edges to a directed acyclic graph are also studied [23]. Furthermore, epidemic spreading processes have been investigated in multiplex networks by considering the impact of heterogeneity and awareness [24].

Except for adaptive strategies [25], the inherent properties of diseases also play a key role in its incidence. For example, it is ubiquitous that some agents are infectious yet indistinguishable in epidemic season since symptoms are not presented (referred as the exposed state). In this sense, an agent is healthy or not just on the basis of symptoms. That is to say, an individual is infected if and only if corresponding symptoms are presented or being validated by conducting some medical tests. Correspondingly, the above-mentioned adaptive frameworks may be confronted with novel challenges (especially for their efficacy) if the exposed state is incorporated. Moreover, the adaptive strategy may excessively destroy the connectivity or functionality to prevent spreading in networks [26], [27].

To resolve this issue, two-layered adaptive multiplex networks are proposed: the physical layer, in particular, supports the spreading process with exposed state (referred as susceptible–exposed–infected–recovered (SEIR) model). In details, the rewiring strategies are conducted as follows:

a susceptible/exposed individual will first disconnect with its infected neighbors probabilistically; once creating a new connection, the selected agent can be in any states except for the infected one. By means of Monte Carlo (MC) simulations, we also study the effects of community movement via adding a number of susceptible individuals into the community.

Along this framework, the contributions of this brief are summarized from two aspects: 1) we incorporate the exposed state into the epidemic spreading process and derive corresponding state transition trees; 2) we apply the proposed SEIR model into an adaptive network and investigate the spreading process in a varying community.

The rest of this brief is organized as follows. The transition probabilities for state transitions are derived in Section II; then the adaptive network with revised rewiring strategy is provided in Section III. Various benchmarks are analyzed in Section IV. Finally, Section V concludes this brief.

II. METHODS

Two-layered multiplex networks are proposed to analyze the epidemic spreading process. Each individual on one-layer maps to its proxy on another layer. In particular, information spreads through various virtual relationships on the virtual layer; while biological elements carrying the virus are exchanged through body contacts on the physical layer.

With regard to spreading processes, an individual on the physical layer falls into one of four states including *susceptible* (*S*), *exposed* (*E*), *infected* (*I*) and *recovered* (*R*). For the *E* state, the viruses are carried and can be transmitted among individuals; however, there is no symptom. On the virtual layer, each agent is either *unaware* (*U*) or *aware* (*A*). The individual will be aware (i.e., $U \rightarrow A$) due to the information spreading process; on the contrary, an aware individual is also likely to forget the awareness, i.e., $A \rightarrow U$, which occurs with probability δ . An aware and susceptible individual takes preventive measures which reduce the transition probability from *S* to *E* by a factor γ_{SE} accordingly. If an individual in *E* is aware, the transition probability from *E* to *I* is reduced by a factor γ_{EI} . For simplicity, we assume $\gamma_{SE} = \gamma_{EI} = \gamma$. For an individual in *E*, the recovery probability is set to be $\hat{\mu}$ if the individual is aware; nevertheless, the recovery rate is reduced by a factor γ_0 . If the scenarios of different values are considered, analyses can be conducted with minor revisions. In total, there will be the following state combinations in networks, i.e., *US*, *UE*, *UR*, *AS*, *AE*, *AI* and *AR*. The corresponding state probabilities for individual *i* are represented by $p_i^{US}, p_i^{UE}, p_i^{UR}, p_i^{AS}, p_i^{AE}, p_i^{AI}$ and p_i^{AR} respectively (*UI* is impractical since an infected individual becomes aware immediately).

For node x_i^j ($j = \{P, V\}$) with m neighbors, once it is in *A* or *E*, any neighbors might be the possible infection source (either epidemic information or disease itself). Here, *P* and *V* indicate the physical layer and virtual one respectively. Correspondingly, the probability of x_i^V not being informed by any aware neighbors (i.e., r_i) is given by $\prod_{l=1}^m (1 - \lambda p_l^A)$, where p_l^A represents the probability of individual *l* being aware and λ represents the information transition probability. Moreover, the probability of an susceptible, aware individual (represented by q_i^A) not being affected by any infectious neighbors (either in *E* or *I*) is derived as $\prod_{l=1}^m (1 - \gamma(\beta' p_l^E + \beta p_l^I))$, where β' and β indicate the probabilities being affected by an exposed and infected individuals respectively. For simplicity, we assume $p_l^{UE} = p_l^{UE} + p_l^{AE}$ and $\beta' = \beta$. Furthermore, the probability of not being affected by any infectious neighbors for

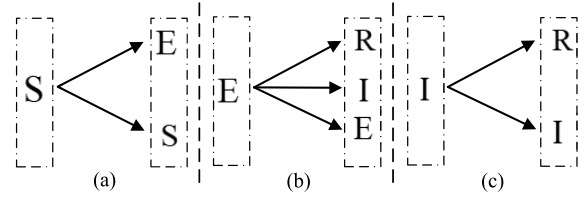


Fig. 1. Transition probability trees for different states.

a susceptible, unaware individual (i.e., q_i^U) is calculated as $\prod_{l=1}^m (1 - (\beta' p_l^E + \beta p_l^I))$ (Above definitions are applicable to the rest of this brief).

The possible transitions from *S* are presented in Fig. 1. Nevertheless, the transition probability is affected by the states of corresponding nodes on the virtual layer. As we see in Fig. 1(a), transition probability from *S* to *E* equals to $1 - q_i^U$ for an unaware individual; otherwise, the state keeps constant with q_i^U . For aware individuals, the transition probability from *S* to *E* is $1 - q_i^A$; otherwise, the state remains unchanged with q_i^A . As in Fig. 1(b), the individual in *E* recovers with certain probability due to self-healing directly; if not recovered, it is infected with certain probability; otherwise, the state remains constant. Corresponding transition probability varies if the individual is aware or unaware of the epidemic. This will be presented later. As in Fig. 1(c), an infected individual is recovered with probability μ ; otherwise, corresponding state does not change.

With provided parameters, corresponding transition probability trees are derived in Fig. 2. Here, the updating process is divided into two sub-steps, i.e., updating process on the virtual layer and updating process on the physical one.

1) *Updating Process on the Virtual Layer*: As Fig. 2(a), for *US*, individual *i* will be informed with probability $1 - r_i$. Hence, the transition from *US* to *AS* occurs with probability $1 - r_i$; otherwise, the state remains constant with probability r_i . Similar analysis can be performed for *UE* and *UR*. For *AE*, the individual is likely to be unaware with probability δ ; otherwise, corresponding state remains constant. This might be incurred by the seasonal property and the fact of forgetting the awareness. Similar analysis can be conducted for *AR* and *AS*. For *AI*, it is impractical for the individual to be unaware once infected.

2) *Updating Process on the Physical Layer*: For *AS*, the individual x_i^P is already aware. That is to say, the transition from *AS* to *AE* occurs with probability $1 - q_i^A$; otherwise, corresponding state combination remains constant with probability q_i^A . Similarly, for *US*, the individual will be exposed with probability $1 - q_i^U$, whereas *US* remains with probability q_i^U . For *AI*, the individual recovers with probability μ due to medical treatment or self-healing; otherwise, the individual remains to be infected.

For *AE*, the individual recovers with probability $\hat{\mu}$ due to self-healing while it can be infected with probability γp . Otherwise, the state remains constant with probability $1 - \hat{\mu} - \gamma p$, where p represents the transition probability from *E* to *I* and $\hat{\mu}$ represent the recovery probability for an individual in *E* if aware. Similar analysis can be performed for *UE*. Nevertheless, *UI* transforms to *AI* immediately as the infected individual is aware timely. Moreover, for *UR* and *AR*, corresponding recovery states remain constant.

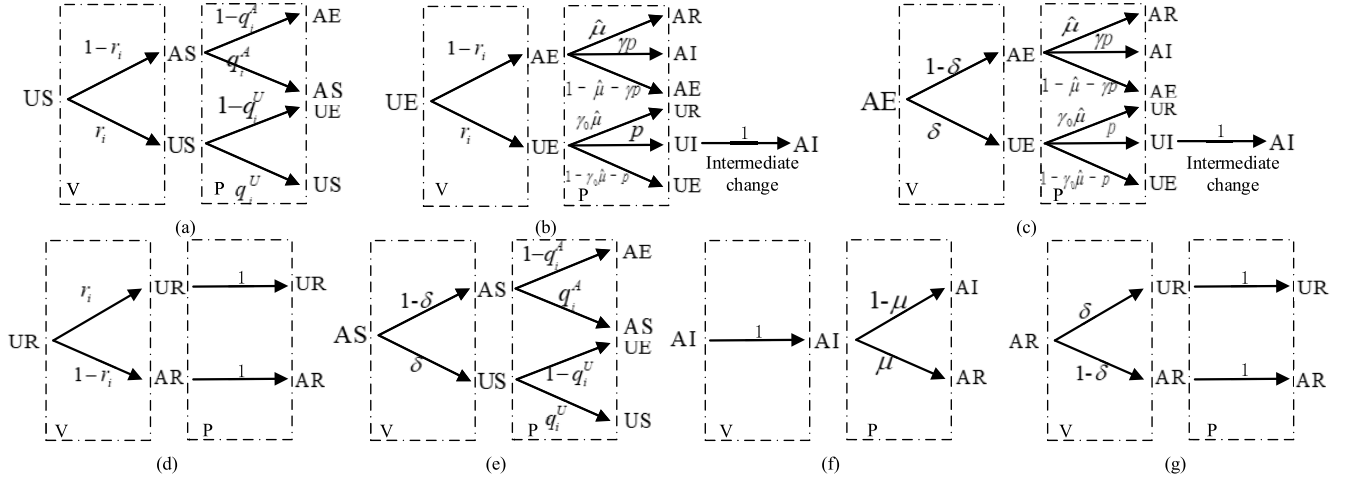


Fig. 2. Transition probability trees for state combination of an individual per time step (a) US, (b) UE, (c) AE, (d) UR, (e) AS, (f) AI and (g) AR.

The probabilities of different state combinations in Fig. 2 for individual i are presented as

$$p_i^{US}(t+1) = p_i^{US}(t) * r_i * q_i^U + p_i^{AS}(t) * q_i^U * \delta, \quad (1)$$

$$p_i^{UE}(t+1) = p_i^{US}(t) * r_i * (1 - q_i^U) + p_i^{AS}(t) * (1 - q_i^U) * \delta + p_i^{UE}(t) * r_i * (1 - \gamma_0 \hat{\mu} - p) + p_i^{AE}(t) * \delta * (1 - \gamma_0 \hat{\mu} - p), \quad (2)$$

$$p_i^{UR}(t+1) = p_i^{UR}(t) * r_i + p_i^{AR}(t) * \delta + p_i^{UE}(t) * \gamma_0 \hat{\mu} * r_i + p_i^{AE}(t) * \delta * \gamma_0 \hat{\mu}, \quad (3)$$

$$p_i^{AS}(t+1) = p_i^{US}(t) * (1 - r_i) * q_i^A + p_i^{AS}(t) * q_i^A * (1 - \delta), \quad (4)$$

$$p_i^{AE}(t+1) = p_i^{US}(t) * (1 - r_i) * (1 - q_i^A) + p_i^{AE}(t) * (1 - \hat{\mu} - \gamma p) * (1 - \delta) + p_i^{UE}(t) * (1 - \hat{\mu} - \gamma p) * (1 - r_i) + p_i^{AS}(t) * (1 - q_i^A) * (1 - \delta), \quad (5)$$

$$p_i^{AI}(t+1) = p_i^{UE}(t) * ((1 - r_i) \gamma + r_i) p + p_i^{AE}(t) * ((1 - \delta) \gamma + \delta) p + p_i^{AI}(t) * (1 - \mu), \quad (6)$$

$$p_i^{AR}(t+1) = p_i^{UR}(t) * (1 - r_i) + p_i^{AI}(t) * \mu + p_i^{AR}(t) * (1 - \delta) + p_i^{UE}(t) * \hat{\mu} * (1 - r_i) + p_i^{AE}(t) * \hat{\mu} * (1 - \delta). \quad (7)$$

Let $p_i^S(t)$, $p_i^E(t)$, $p_i^I(t)$ and $p_i^R(t)$ represent the probabilities for individual i in S , E , I and R at time t ($p_i^S(t) + p_i^E(t) + p_i^I(t) + p_i^R(t) = 1$) respectively. For corresponding node on the virtual layer, probabilities of being aware or unaware are defined as $p_i^U(t)$, $p_i^A(t)$ respectively (here, $p_i^U(t) + p_i^A(t) = 1$). Hence, theoretical (Markov chain) analyses can be performed using the above formulas.

In practice, it takes some time (specified as t_m) for observing symptoms. Thereby, the transition process in Fig. 1(b) is revised as that in Fig. 3. Once an individual is in E , the time flag t_u is set to be 0. At each time step, the model in Fig. 3 is adopted and t_u is increased by 1. This process repeats until t_u reaches t_m . Then, the exposed individuals switch to I mimicking the fact that symptoms usually appear definitely after some time.

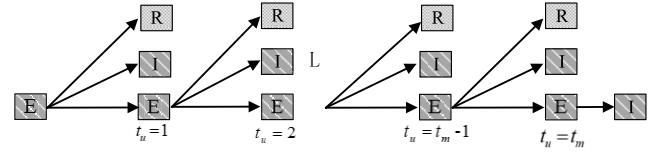


Fig. 3. Transition process for an individual in E .

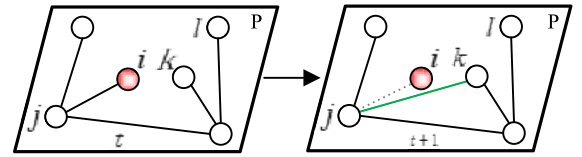


Fig. 4. Illustration of the rewiring strategy. Red and white nodes indicate the infected and uninfected cases respectively.

III. REVISED ADAPTIVE NETWORKS

In this brief, the physical layer is assumed to be adaptive: an individual can avoid contacting with infected rather than exposed neighbors as symptoms are not presented. In details, the link rewiring strategy is as follows: the susceptible/exposed individuals are likely to disconnect their infected neighbors for self-protection with a certain probability (i.e., q_0). Once disconnected, a new connection will be constructed with another individual definitely which is randomly selected from uninfected agents (i.e., S , E , R) rather than existing neighbors. Here, the newly selected individual can be in any states except I . Moreover, the isolation avoidance is considered, that is to say each node has at least one neighbor.

For an illustration, Fig. 4 shows the revised link rewiring strategy. In Fig. 4, i is infected and the connection between i and j should be removed with certain probability. Once deleted, a new connection will be constructed with either k or l (here, k is chosen for an illustration; nevertheless, whether viruses are carried or not is unknown due to the lack of symptoms).

Furthermore, the effects of incorporating new individuals are also investigated. At each time step, N_n susceptible individuals are newly added; while each new individual is randomly connected with N_l individuals in S , E , or R .

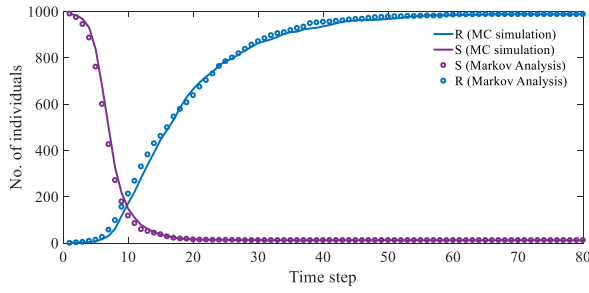


Fig. 5. Comparison of the results using different approaches for the scenario without t_m and $N_n = 0$.

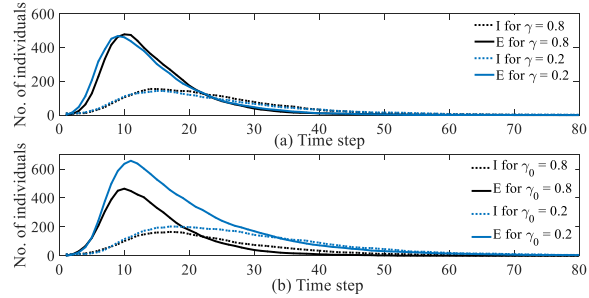


Fig. 6. (a) Numbers of exposed and infected individuals for (a) $\gamma = 0.2$ and 0.8. (b) $\gamma_0 = 0.2$ and 0.8. Here, the scenario without t_m and $N_n = 0$ is considered.

IV. RESULTS ANALYSIS

For simplicity, a Barabasi-Albert (BA) network consisting of $N = 1000$ nodes on each layer is used for analysis ($< k > = 8$). The other parameters are listed as $p = 0.1$, $\lambda = 0.02$, $\hat{\mu} = \mu = 0.1$, $\gamma = 0.8$, $\beta = \beta' = 0.12$, $\gamma_0 = 0.8$ and $\delta = 0.2$. 10 individuals are randomly chosen to be infected initially. That is to say, corresponding nodes on the virtual layer for the infected individuals become aware immediately. Monte Carlo (MC) simulation is conducted, while it stops if stationary states or anticipated time steps are achieved.

Simulations are conducted and averaged over 50 trials with corresponding simulation-based results being presented in Fig. 5 indicated by solid lines; whereas the results through Markov chain analyses are denoted by solid points. It is clear that the results of simulation-based and Markov chain analyses coincide well. With the increment of time step, the number of susceptible individuals reduces first and eventually stabilizes around certain value, whereas the number of recovered individuals increases first and stabilizes eventually.

To reflect the effects of varying γ or γ_0 , different scenarios are considered with the results being presented in Fig. 6 respectively. We see that, the number of exposed individuals increases first and then reduces; similar trend exists for the infected ones. As indicated, the peak of exposed/infected individuals is higher if a larger value of γ is adopted. This is due to the reason that a small factor γ indicates the transition probability from S to E (or from E to I) will be reduced by a large extent if preventive measures are taken. Nevertheless, for γ_0 , if a smaller value is adopted, then it is less likely for the exposed/infected individuals to be recovered. Hence, a smaller γ_0 indicates a higher summit.

To illustrate the effect of adaptive behavior on the range of individuals being affected, two different rewiring strategies are considered (Scenario 1: only S individual destroys

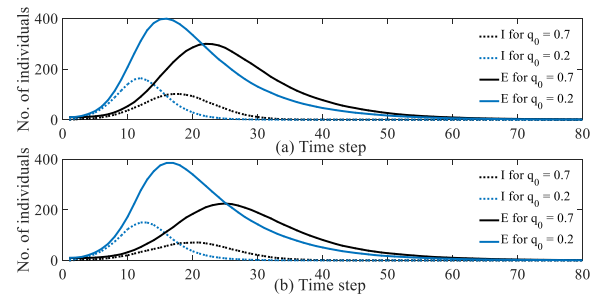


Fig. 7. Illustration of the number of individuals being affected for the scenarios without t_m and $N_n = 0$ if $q_0 = 0.2, 0.7$ (Here, $p = 0.3$).

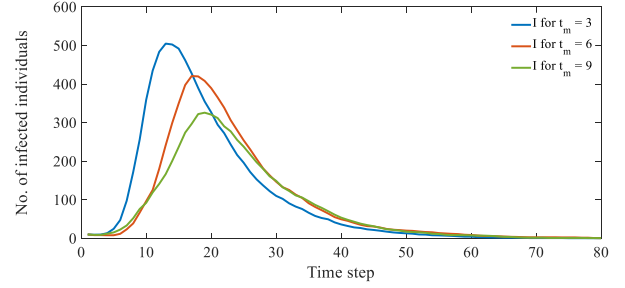


Fig. 8. Relationships between the number of infected individuals and different t_m for the scenario of $N_n = 20$ with $N_l = 3$, $q_0 = 0.2$.

the connection with an infected neighbor; Scenario 2: an individual in S/E can disconnect the connection) in Fig. 7(a) and (b) respectively. As in Fig. 7(a), the spreading of epidemic can be postponed for both strategies and with a smaller peak as q_0 increases. Furthermore, the range of affected individuals is reduced for larger q_0 in Scenario 1. If q_0 is larger, connections with infected individuals will be more likely to be cut. Hence, it is hard for the spreading of epidemics. In particular, compared with the first scenario, Fig. 7(b) shows lower infection peak and incidence (black lines for large q_0). This is because both S and E nodes will cut connections with infection sources more effectively, which prevents larger outbreak. While for lower q_0 , there is no great difference of infection incidence between two cases, since the total amount and used time of cutting infection neighbors is nearly the same.

For different simulations, the parameters also play important roles in affecting the spreading process. Here, the effects of varying parameters on the proportion of focused variables are investigated. For saving space yet without losing generality, the relationships between the number of individuals being affected and different t_m are illustrated in Fig. 8. It becomes harder for the community to be affected and the summit can be postponed with the increase of t_m . Thus, for larger t_m , it is more likely for the exposed individuals to be in state E and an exposed individual will be less likely to directly recovered.

Furthermore, the scenario of varying community ($N_n = 20$ and $N_l = 3$) is also investigated with results being presented in Fig. 9. It is clear that λ plays an important role in affecting the epidemic spreading process for the provided parameters. As indicated, the number of individuals being affected reduces with the increase of λ . This is because that the epidemic information will spread rapidly if larger λ is adopted; once a susceptible individual is aware, preventive measures can be taken which reduces the probability of being affected by the epidemic. If the effect of awareness being forgotten is considered, analyses can be conducted accordingly.

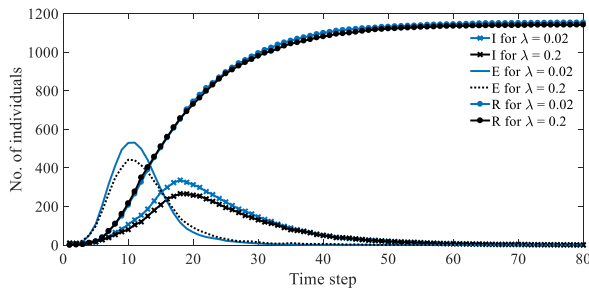


Fig. 9. Relationships between the number of individuals and λ for Scenario 2. $t_m = 9$ while the other parameters are the same as those for Fig. 8.

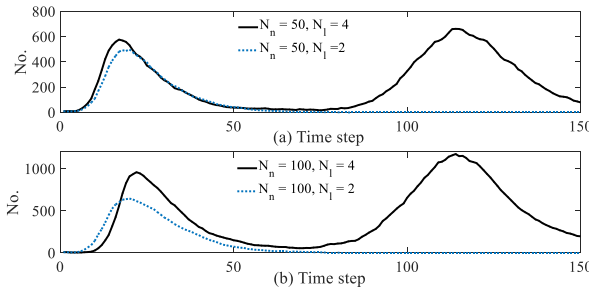


Fig. 10. Numbers of infected individuals for different N_n and N_l .

Furthermore, the effects of varying N_n and N_l on the epidemic are illustrated in Fig. 10. The epidemic is likely to reoccur for large N_n or N_l . This indicates that the epidemic is unlikely to be eliminated completely and the connections with exposed/infected individuals cannot be totally disconnected. The newly added individuals are more likely to connect with unaffected individuals. Thus, these newly added individuals can form a community of susceptible and recovered individuals. For larger N_n or N_l , the possibility of connecting with newly added individuals increases. Once the infectious individual connects with an individual in the healthy community; the epidemic is likely to re-burst. For smaller N_n or N_l , such possibility reduces. Thus, N_n and N_l play important roles in epidemic spreading.

V. CONCLUSION

In practice, most networks exhibit layered properties which are characterized as multiplex networks. The epidemic spreading processes in adaptive multiplex networks are proposed: facing undesirable infections, individuals will take preventive measures to avoid direct contacts with infected neighbors. In particular, an exposed state is incorporated, and corresponding state transition trees can be derived. Then, theoretical analyses through Markov chain can be performed. Nevertheless, with the incorporation of exposed state, it is difficult to discriminate the exposed neighbors from infected ones. Hence, a revised rewiring strategy is presented. Then, the dynamics of epidemic spreading on the aforementioned adaptive network are thoroughly investigated. The importance of considering virtual layer is illustrated, such as, the infectivity probability and recovery probability, once corresponding individuals are aware. It is unveiled that the incorporation of adaptive strategy plays an important in suppressing the epidemic spreading process. Moreover, the effects of time periods required for the transition from E to I are also studied. It is also found that the incorporation of community movement plays an important role in the epidemic re-occurrence.

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